

BIRDS

PHYLUM	Chordata
CLASS	Aves
ORDERS	40
FAMILIES	240
SPECIES	10,797

Birds are the most accomplished of all flying animals. Their ability to fly has allowed them to spread throughout the world, often to places, such as remote islands and

Antarctica, that are beyond the reach of many other animals. Like mammals, birds are endothermic (warm-blooded) vertebrates. However, unlike most mammals, they reproduce by laying eggs. Birds have several adaptations for flight, including wings, feathers, a light but strong skeleton, and a highly efficient respiratory system.

Evolution

Birds evolved from reptilelike ancestors, possibly from tree-dwelling dinosaurs that fed on insects. A lifestyle of arboreal hunting would have promoted the development of such birdlike characteristics as large eyes, grasping feet, and a long snout, later to evolve into a bill. It might also have resulted in the transition from being cold- to warm-blooded—an

advantage for animals that feed on insects, which become inactive and slow-moving in cold conditions. Feathers, which were derived from reptilian scales, probably first evolved to provide insulation, although they were no doubt put to the purpose of flight very early on.

One of the earliest-known bird fossils is about 150 million years old, dating from the Jurassic Period (205–142 million years ago). Named *Archaeopteryx lithographica*, this animal was about the size of a crow and showed a combination of reptilian and avian features: it had wings and feathers like a bird but also had a snout, rather than a bill, and the toothed jaws of a reptile. There is some doubt about whether



TAKING OFF
The power of flight has enabled birds to exploit aerial niches. In many habitats, birds of prey (such as this tawny eagle), with their powerful wings, acute vision, and sharp bill and talons, are the top predators.

flight surface (see panel, below, and p.266); they also provide protection and insulation.

The skeleton of a bird combines remarkable lightness with strength, attributes that are essential for powerful flight. To restrict the bird's



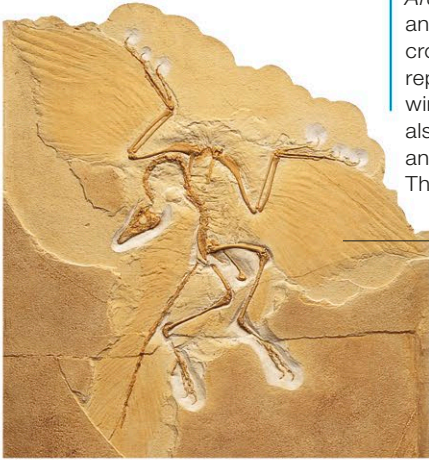
CROSS SECTION OF A BIRD'S BONE
Many bones in a bird are hollow, including the major limb bones, parts of the skull, and pelvis. This reduces the bird's weight, to conserve energy for flight. The bones are strengthened by internal struts.

Archaeopteryx could fly or just glide, since it lacked the keeled breastbone that provides attachment sides for the muscles needed for powered flight.

During the Cretaceous Period (145–65 million years ago), birds diversified and their anatomy evolved to make possible increasingly efficient flight. It was in this period that the ancestors of living birds appeared. Toward the end of the Cretaceous, a wave of mass extinctions saw the end of the age of the dinosaurs. It is not clear why birds survived—perhaps being warm-blooded helped them withstand a prevailing climatic catastrophe. But having survived, birds as a group flourished into the diversity of forms alive today.

Anatomy

A bird has several physical adaptations for flight. Its body is short, strong, and compact, with powerful muscles for moving the wings, and strong legs to launch it into the air and cushion the impact of landing. Its feathers form the



impressions made by feathers

ARCHAEOPTERYX FOSSIL

Archaeopteryx lithographica is thought to represent a link between reptiles and birds: the jaws, snout, and tail (which is supported by vertebrae) are reptilian, but the wings and feathers are birdlike.

FEATHERS

Feathers are highly complex structures that are unique to birds. However, they are formed from the same material, keratin, that is found in the hair of mammals and the scales of reptiles. Feathers are subject to considerable wear and tear. To keep them in good condition, birds regularly clean, oil, and reshape their plumage using their bill, an activity known as preening. Other forms of maintenance include scratching, bathing, and sunning. Feathers are shed (molted) and replaced at least once a year. Flying birds have 4 different kinds of feathers, each of which is modified to serve a distinct function: there are 2 types of flight feathers (wing and tail feathers), as well as down feathers and contour feathers.

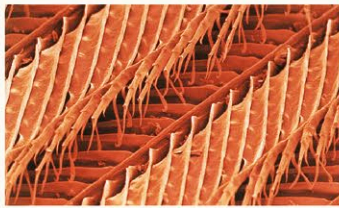


WING FEATHER

The feathers along the edge of the wing are long and rigid with a clearly defined shape, providing the lift for flight and maneuvering. Unlike the tail feathers, which are often symmetrical, wing feathers are unevenly shaped.

FEATHER TYPES

Down feathers form an insulating underlayer, while small contour feathers provide a streamlined covering over the body. The long tail feathers are used for flying and steering.



FEATHER STRUCTURE

The structure of feathers is extremely complex. Most feathers have a central shaft (rachis), from which closely spaced branches (barbs) project outward to form a continuous, flat surface. Minute side branches (barbules) lock the barbs together.